

An aerial, grayscale photograph of a city street scene. A train track runs vertically through the center of the image. On either side of the track are various buildings, some with flat roofs and others with more complex structures. Streets with crosswalks and some vehicles are visible. The overall tone is dark and technical.

Ray Casting

Lecture 2 Fall 2025

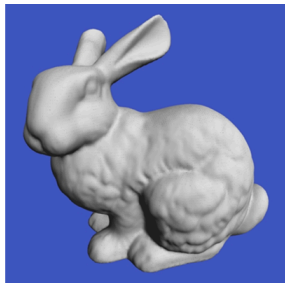
Press Space for next page →

Announcements

- Assignment 1 is due this week (Friday, Sep 12, 11:59 PM)
 - Office hours for A1 online this Friday
- Assignment Grace Tokens:
 - 10 tokens that each provide an additional 12 hours to submit penalty free.
 - Read more on the course Github.

Last Time: Photography

Images captured from the real world.



Object



Light



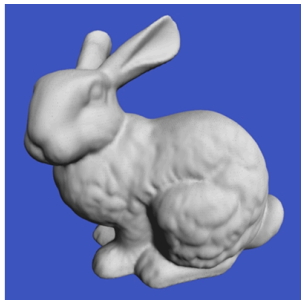
Camera



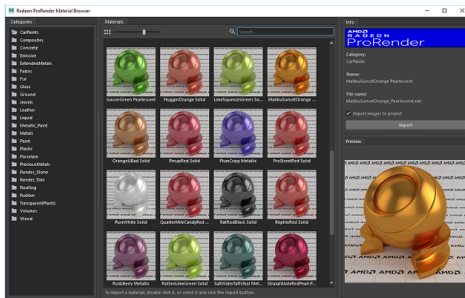
Image

Rendering: From Real to Virtual

How do we generate images using digital scenes?



Object



Materials



Image

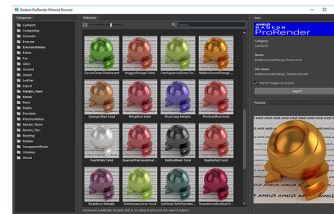
Overview of Rendering

What problem are we actually solving?

Defining the image function $I(x, y)$



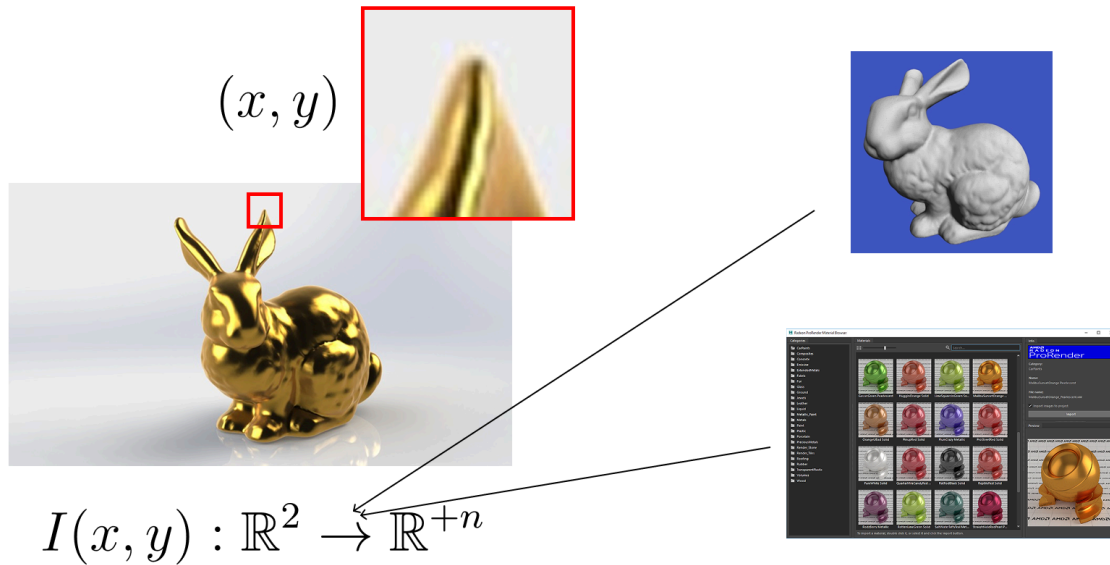
$$I(x, y) : \mathbb{R}^2 \rightarrow \mathbb{R}^{+n}$$



...

What problem are we actually solving?

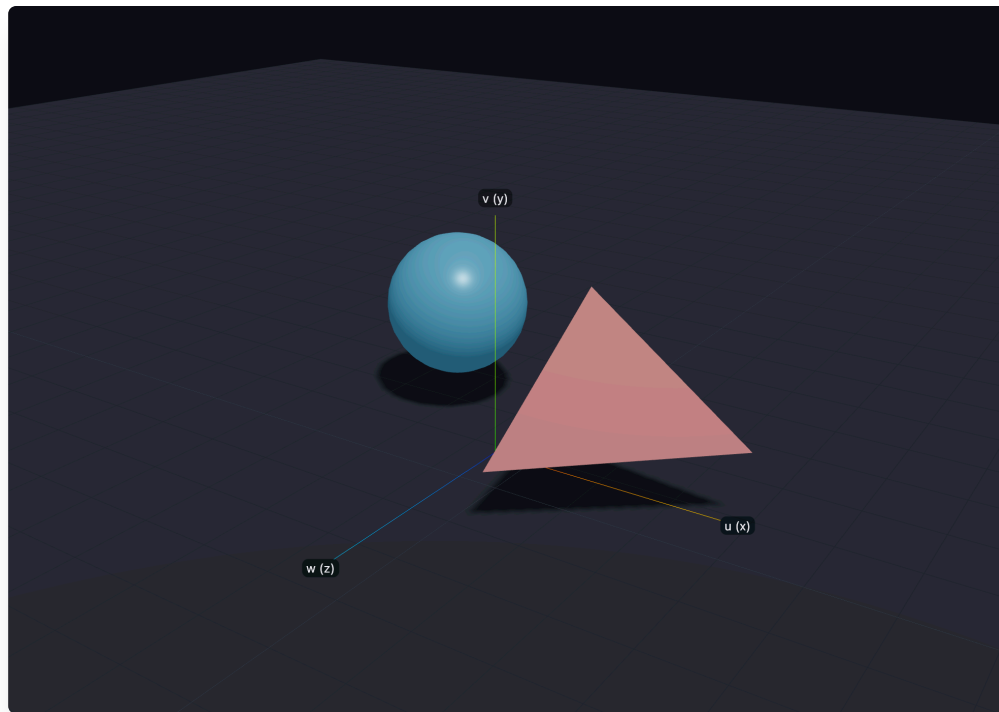
Defining the image function $I(x, y)$



...

3D Scene

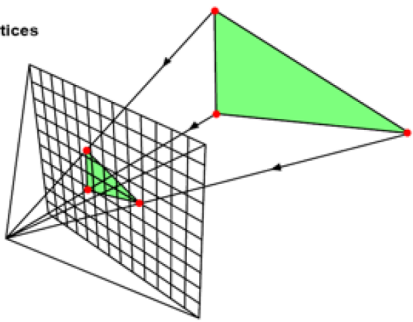
A simple virtual scene with basic geometry



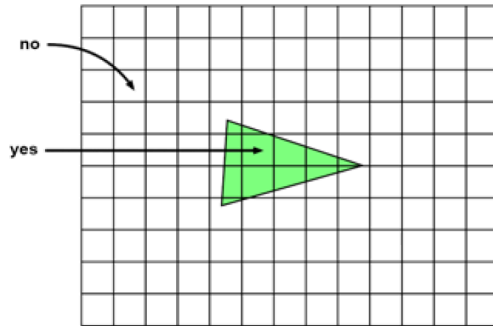
Rasterization

Projecting to the image plane

1) Project vertices

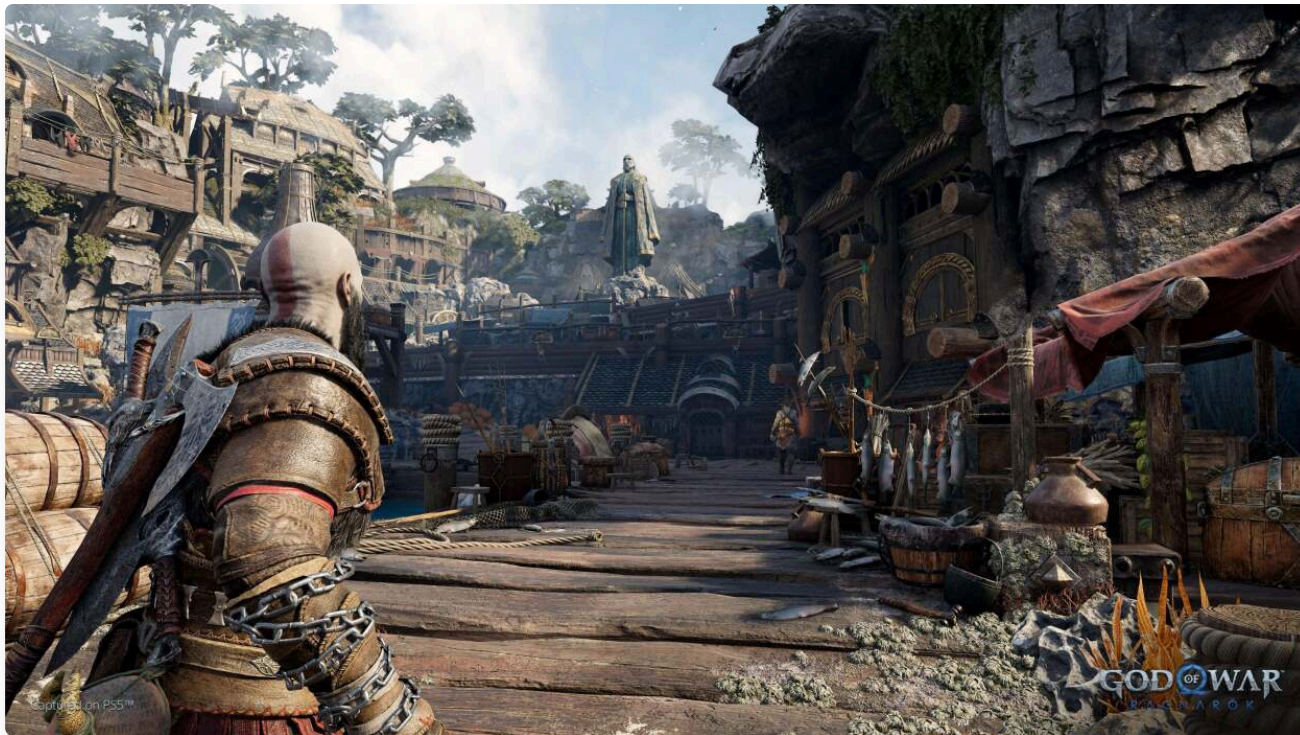


© www.scratchapixel.com



Rasterization

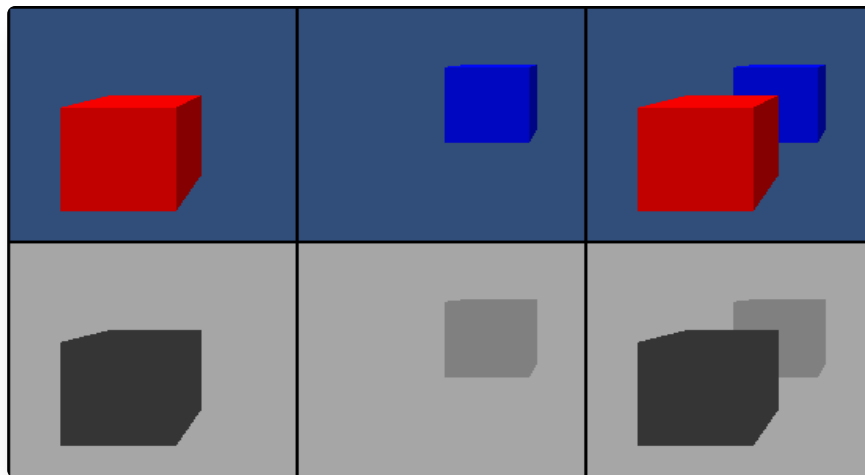
Projecting to the image plane



Rasterization Loop

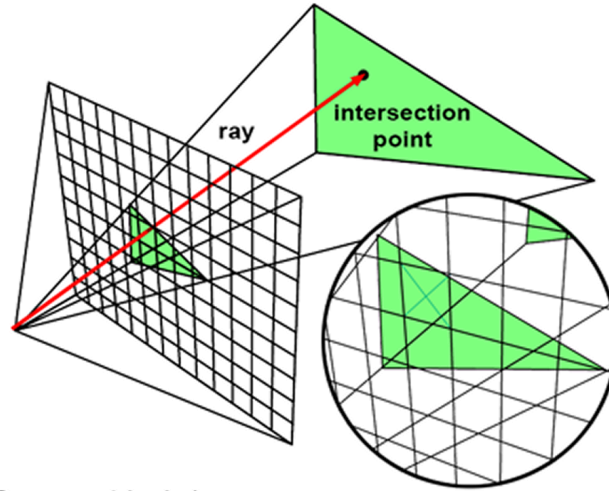
Object-order Rendering: Objects first, then pixels.

```
1  for each scene object {  
2    for each pixel {  
3      if (object affects pixel) {  
4        Do something  
5      }  
6    }  
7  }
```



Ray Casting

Projecting rays from pixels into the scene



© www.scratchapixel.com

Ray Casting

Projecting rays from pixels into the scene



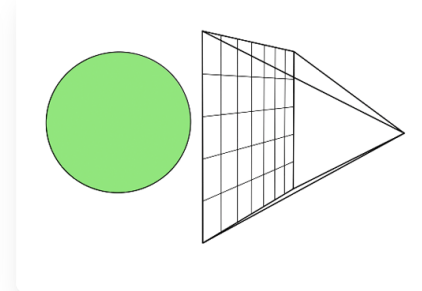
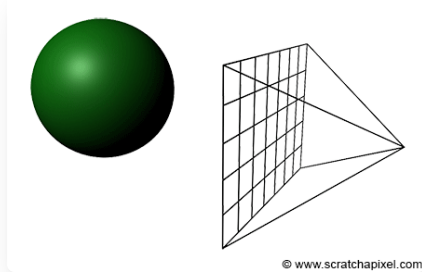
<https://www.scratchapixel.com/lessons/3d-basic-rendering/rasterization-practical-implementation/overview-rasterization-algorithm.html>

<https://computergraphics.stackexchange.com/questions/4585/mirror-reflections-ray-tracing-or-rasterisation>

Ray Casting Loop

Image-order rendering: Pixels first, then objects.

```
1  for each pixel {  
2    generate a ray  
3    for each object {  
4      if (ray intersects object) {  
5        Do something  
6      }  
7    }  
8  }
```



Difference in Loops

Trade-offs

```
for each pixel {  
    generate a ray  
    for each object {  
        if (ray intersects object) {  
            Do something  
        }  
    }  
}
```

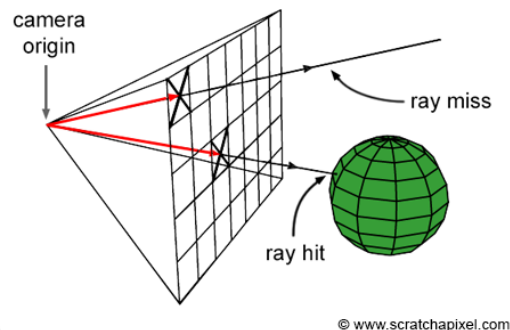
- Faster and Parallel
- Good for realistic lighting
- Works with many geometric representations

```
1  for each scene object {  
2      for each pixel {  
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```

- Excellent for realistic lighting
- Works with many geometric representations (meshes, implicit surfaces, volumes, voxels)
- Slow and noisy

Ray Casting Overview

```
1  for each pixel {  
2    generate a ray  
3    for each object {  
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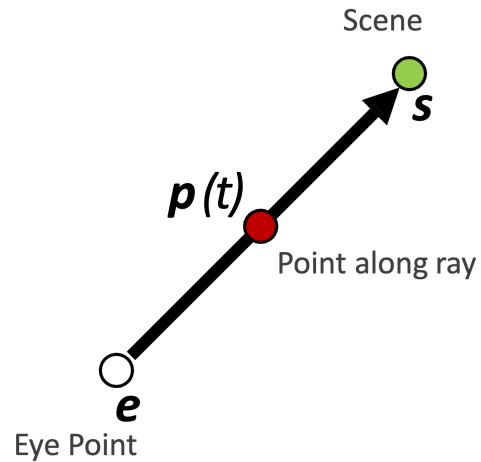
- ☐ What am I shooting?
- ☐ Where is being shot?
- ☐ How do I know if I hit something?

Ray

Origin + Direction

$\vec{p}(t) = \vec{e} + t\vec{d}$ where:

- $\vec{e}$ = origin vector
- $\vec{d}$ = direction vector
- t = time or distance along ray (in steps of $|\vec{d}|$)



Ray Practice

Practice

Question: If I have a point $\vec{a} = (1, 1)$ and a point $\vec{b} = (3, 2)$, define a ray that starts at point $\vec{a}$ and goes through point $\vec{b}$.

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$$(19, 10) = (1, 1) + t(2, 1)$$

$$(19, 10) = (1 + 2t, 1 + t)$$

$$t = 9$$

Ray Practice

Practice

Question: If I have a point $\vec{a} = (1, 1)$ and a point $\vec{b} = (3, 2)$, define a ray that starts at point $\vec{a}$ and goes through point $\vec{b}$.

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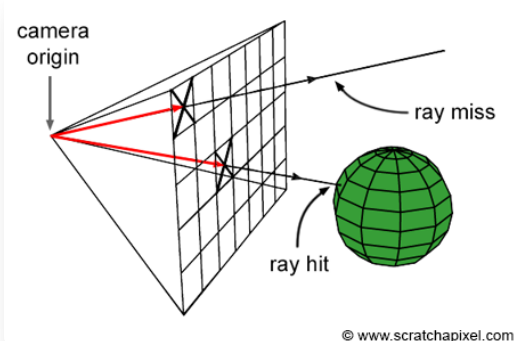
$$(19, 10) = (1 + 2t, 1 + t)$$

$$t = 9$$

**Important: If we have 2 points, we have a way to define a ray.*

Ray Casting Mechanisms

```
1  for each pixel {  
2    generate a Ray  
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```



- ☒ What am I shooting? A Ray: $p(t) = e + t \cdot d$
- ☐ Where is being shot?
- ☐ How do I know if I hit something?

3D scene <-- ? --> 2D pixel grid

What ties the 3D scene and the 2D pixel grid?

The 3D image plane is a conceptual plane that sits between the 3D scene and the 2D pixel grid.

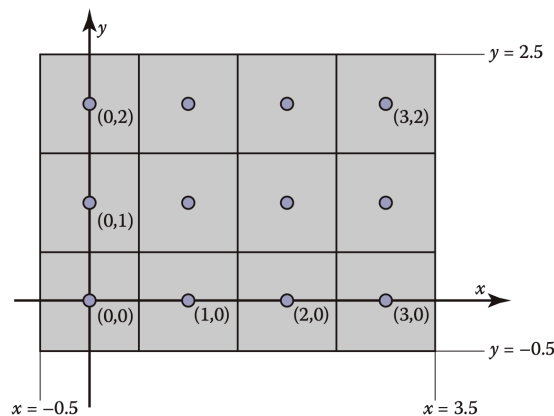
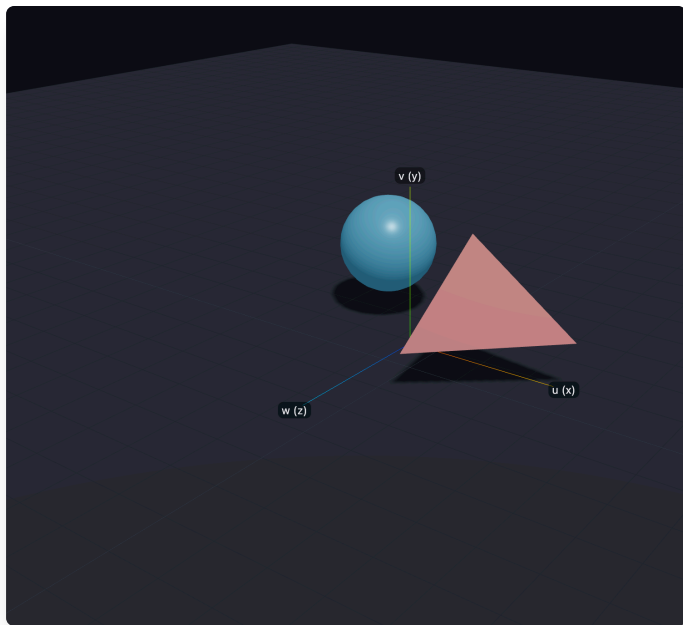


Figure 3.10. Coordinates of a four pixel $\times$ three pixel screen. Note that in some APIs the y -axis will point downward.

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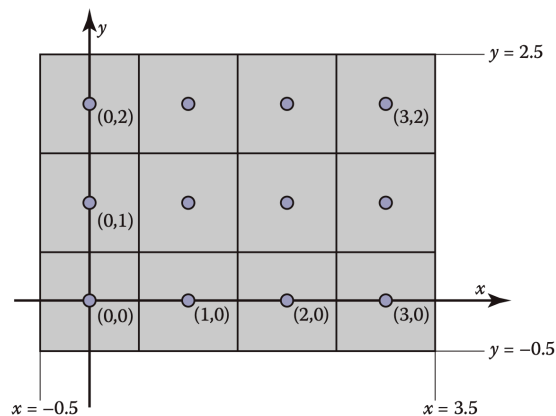
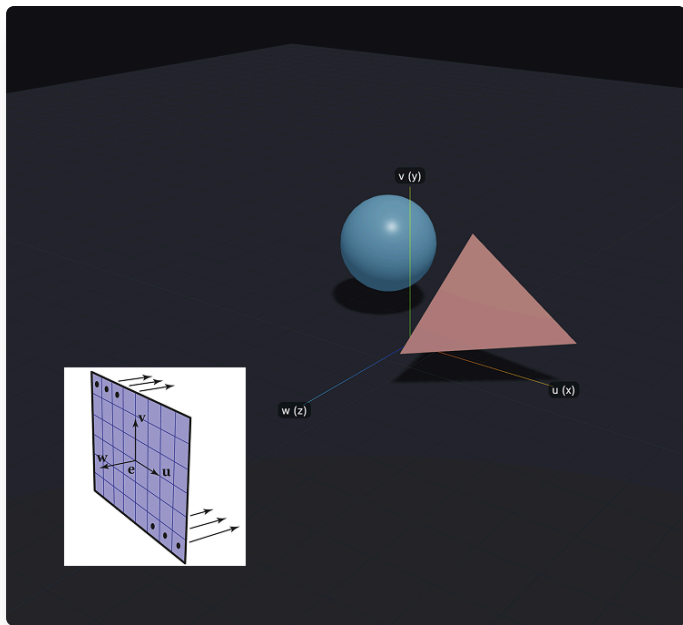
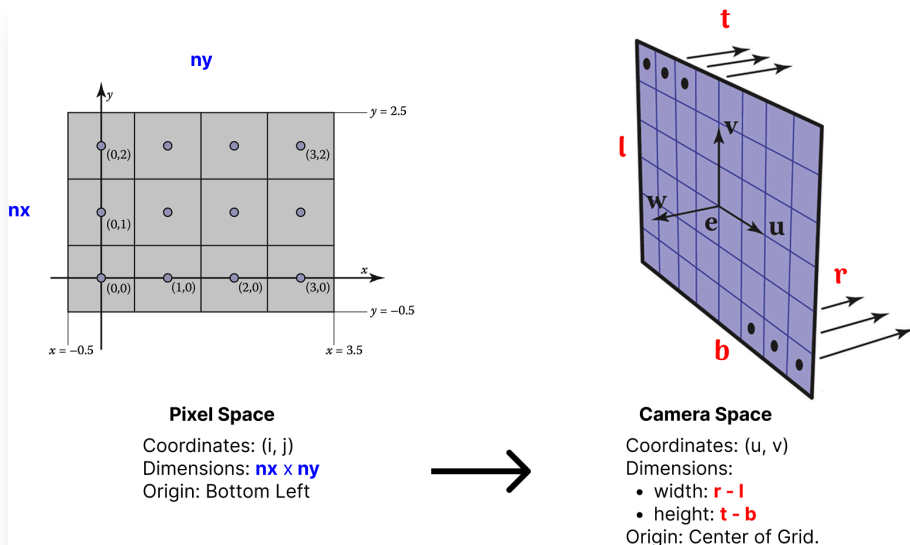


Figure 3.10. Coordinates of a four pixel $\times$ three pixel screen. Note that in some APIs the y -axis will point downward.

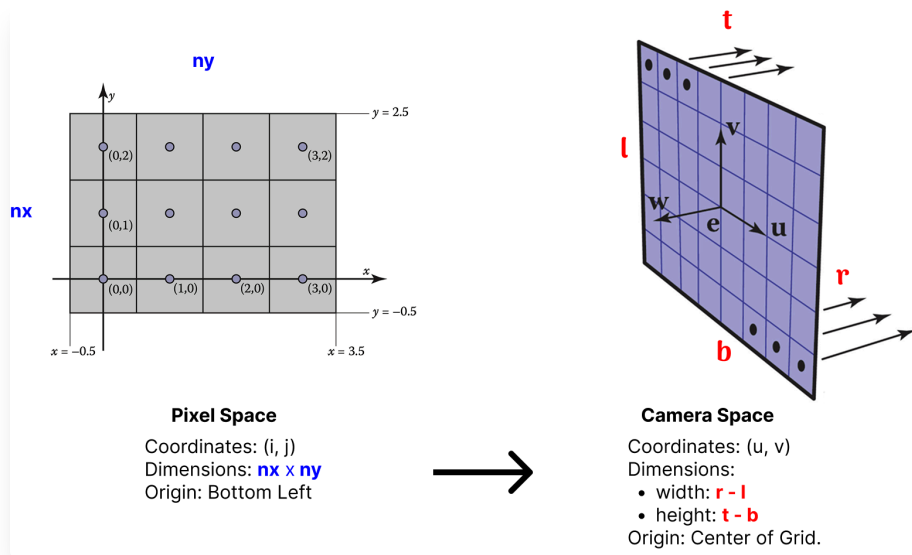
Mapping 2D pixel grid to 3D image plane

How does each pixel (i, j) map to a point (u, v) on the image plane?



Mapping 2D pixel grid to 3D image plane

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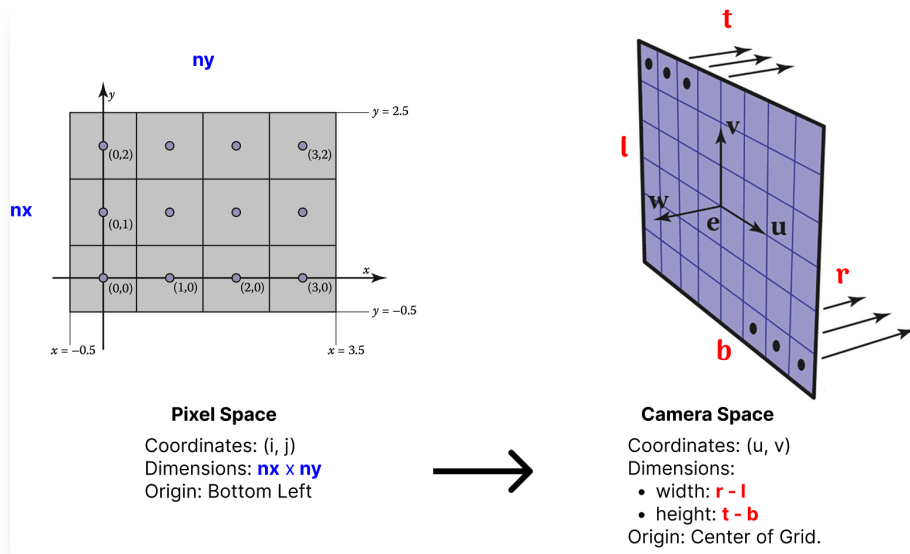


Q: What is the bottom-left corner of the pixel grid? Of the Image Plane?

Q: What is the top-right corner of the pixel grid? Of the Image Plane?

Mapping 2D pixel grid to 3D image plane

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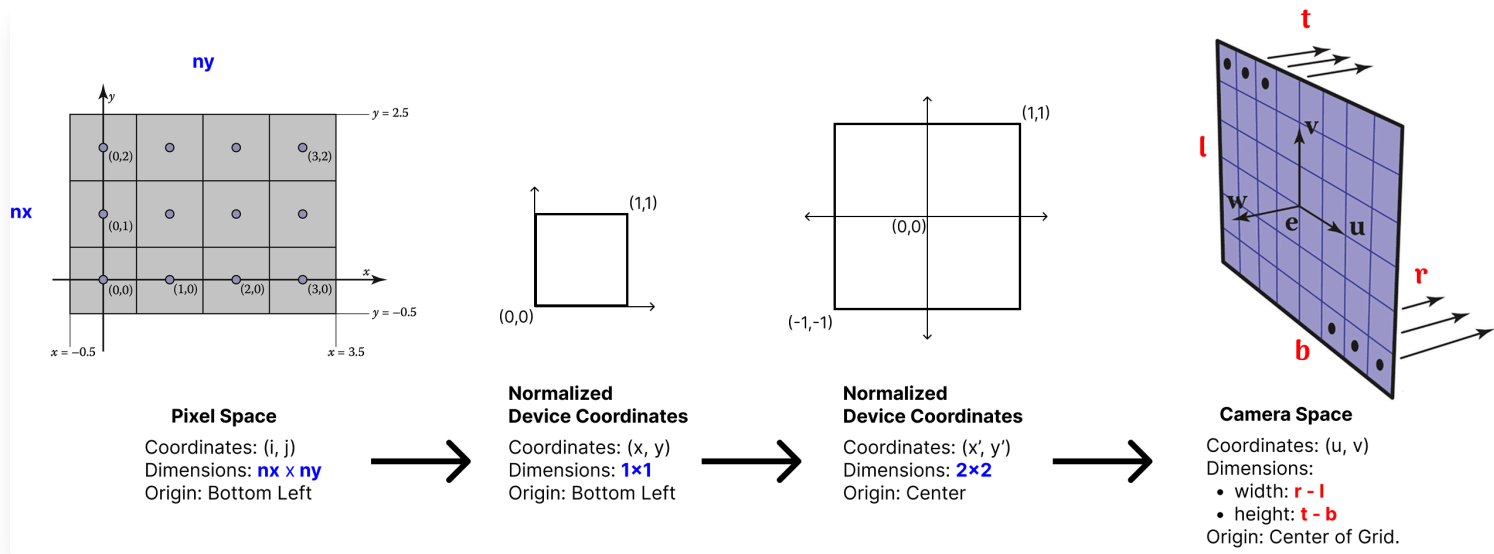


Q: What is the bottom-left corner of the pixel grid? Of the Image Plane? $(-0.5, -0.5)$ and (l, b)

Q: What is the top-right corner of the pixel grid? Of the Image Plane? $(nx - 0.5, ny - 0.5)$ and (r, t)

Mapping 2D pixel grid to 3D image plane

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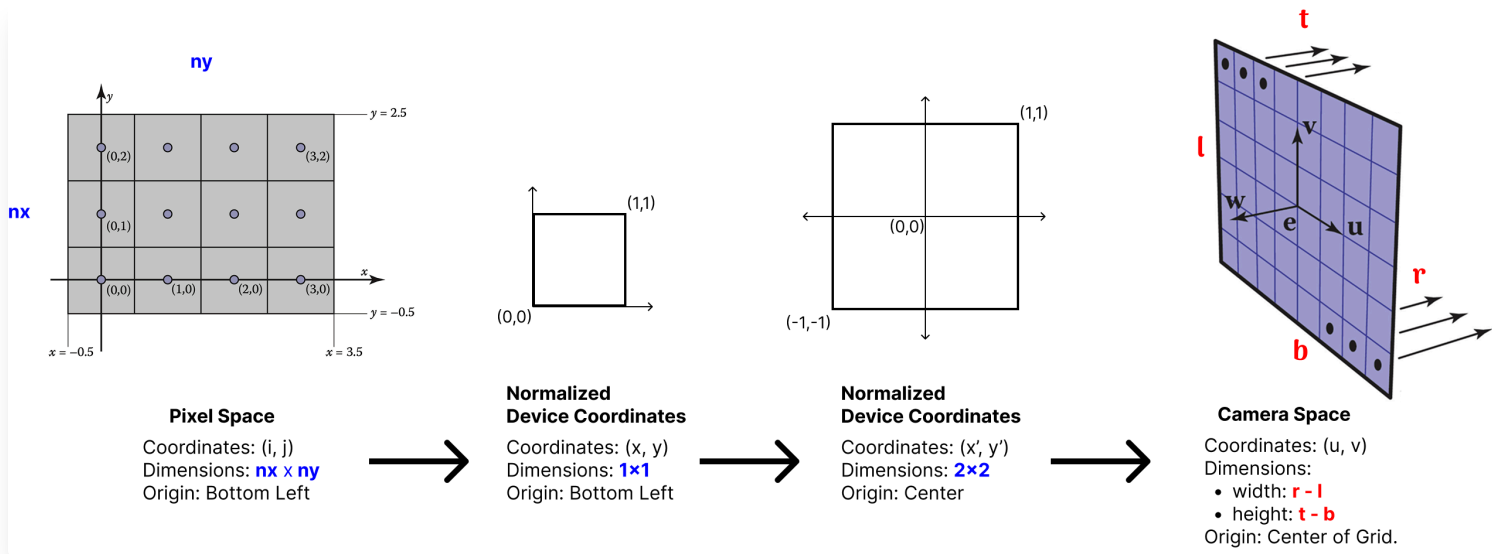


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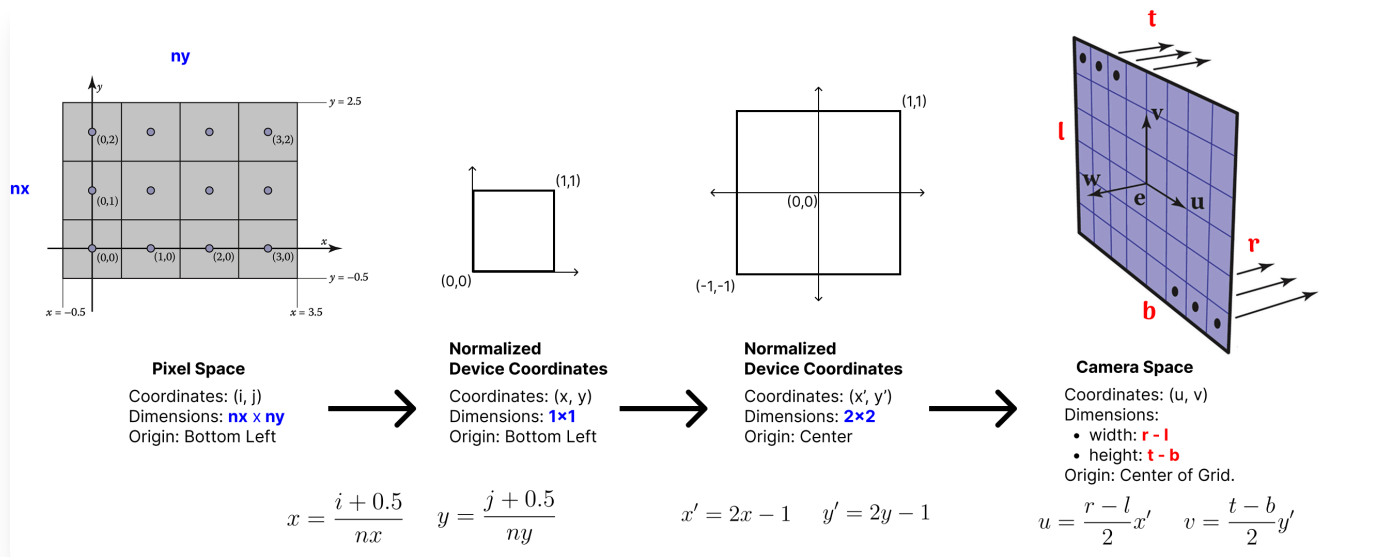
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Q: What is the top-right corner of the pixel grid? Of the Image Plane? $(nx - 0.5, ny - 0.5)$ and (r, t)

Try to derive the mapping from (i, j) to (u, v) .

Mapping 2D pixel grid to 3D image plane

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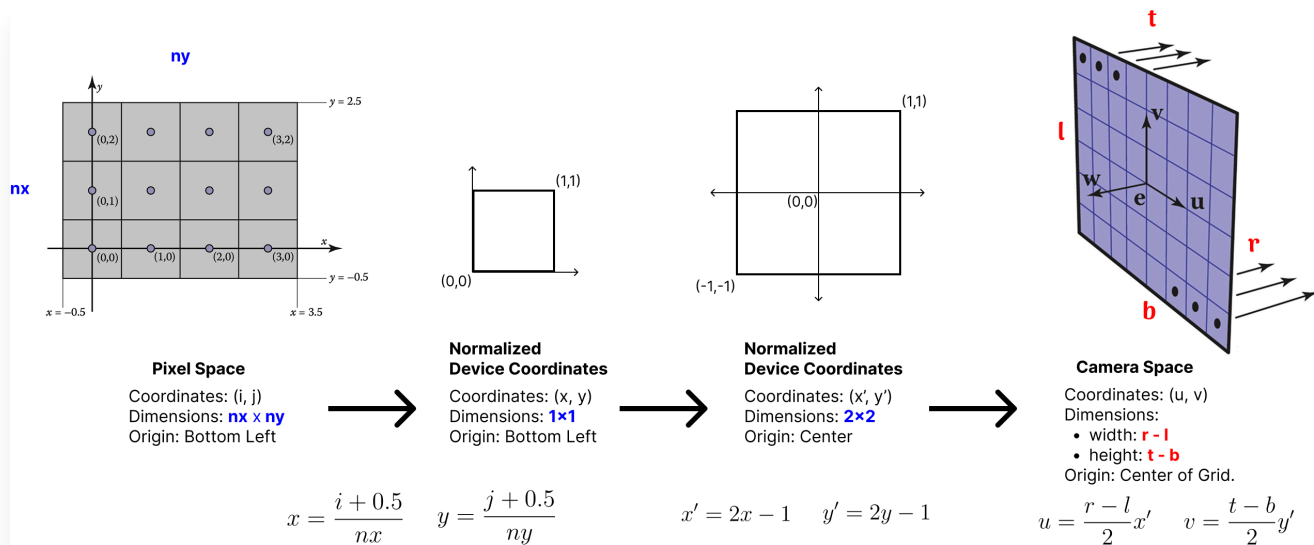
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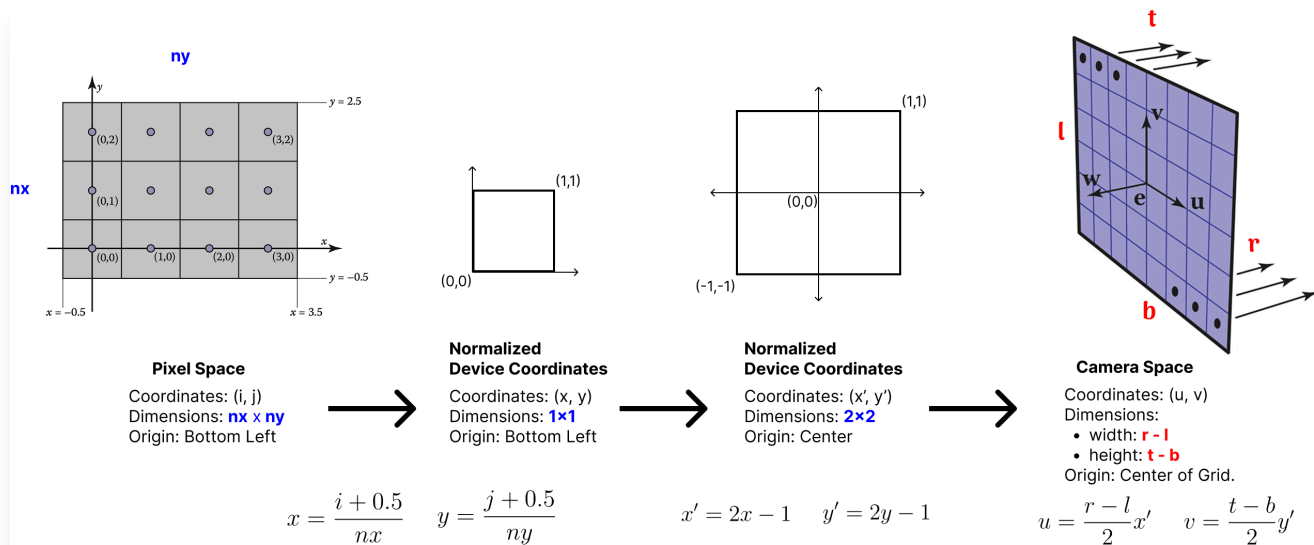


$$u = \frac{(r - l)}{nx} (i + 0.5) - \frac{(r - l)}{2}$$

$$v = \frac{(t - b)}{ny} (j + 0.5) - \frac{(t - b)}{2}$$

Mapping 2D pixel grid to 3D image plane

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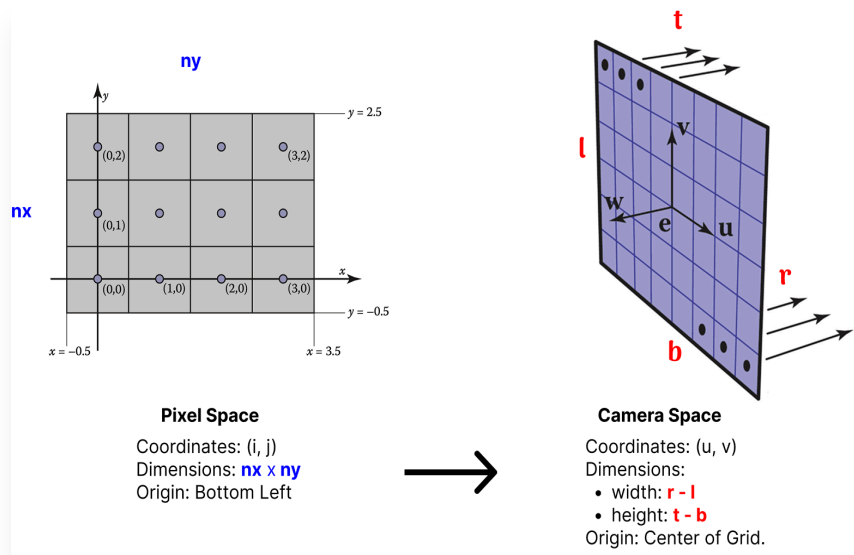


$$u = \frac{width}{nx} (i + 0.5) - \frac{width}{2}$$

$$v = \frac{height}{ny} (j + 0.5) - \frac{height}{2}$$

Shooting rays

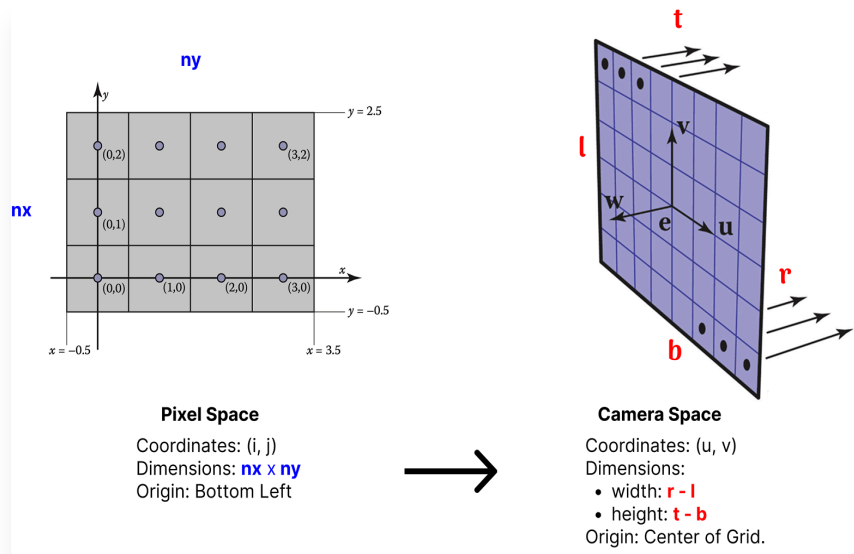
Which direction do we shoot rays from the image plane?



$$u = \frac{width}{nx}(i + 0.5) - \frac{width}{2}$$
$$v = \frac{height}{ny}(j + 0.5) - \frac{height}{2}$$

Shooting rays

Which direction do we shoot rays from the image plane?

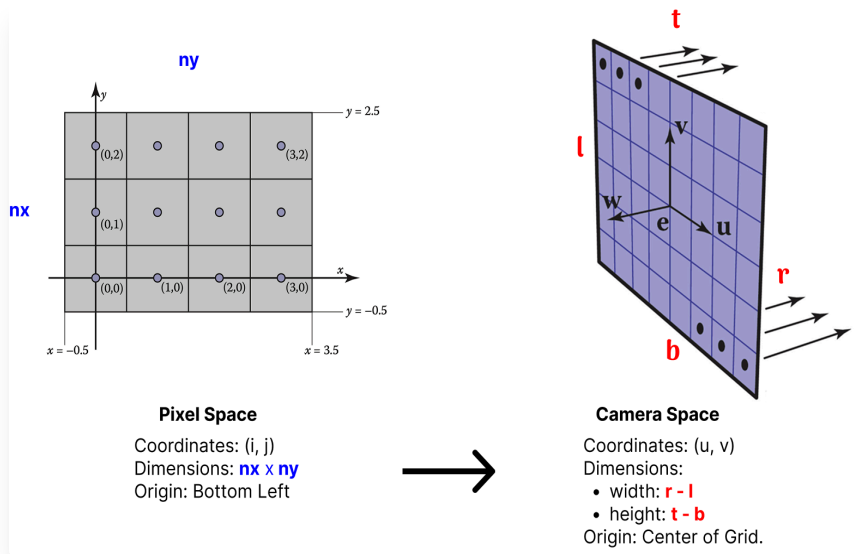


$$\vec{p}(t) = \vec{e} + t\vec{d}$$

$$u = \frac{width}{nx}(i + 0.5) - \frac{width}{2}$$
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Shooting rays

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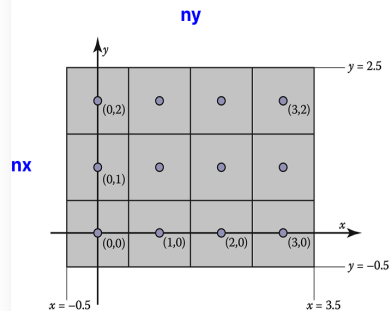
$$\vec{e} = (u, v, 0)$$

$$\vec{d} = -\vec{w}$$

$$u = \frac{width}{nx}(i + 0.5) - \frac{width}{2}$$
$$v = \frac{height}{ny}(j + 0.5) - \frac{height}{2}$$

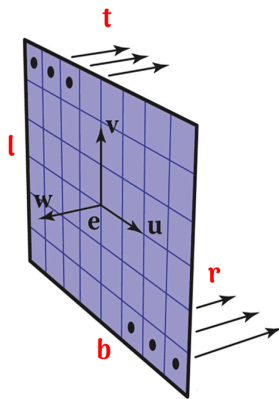
Orthographic Projection

Same direction, different origin.



Pixel Space

Coordinates: (i, j)
Dimensions: $nx \times ny$
Origin: Bottom Left



Camera Space

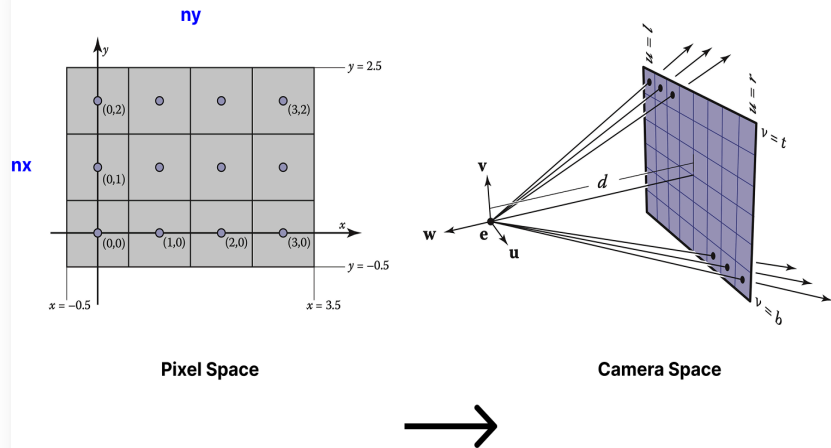
Coordinates: (u, v)
Dimensions:
• width: $r - l$
• height: $t - b$
Origin: Center of Grid.

```
for each pixel {  
    u = width/nx * (i + 0.5) - width/2  
    v = height/ny * (j + 0.5) - height/2  
    e = (u, v, 0)  
    d = -w  
  
    for each object {  
        if (ray intersects object) {  
            Do something  
        }  
    }  
}
```

Demo

Perspective Projection

Same origin, different direction.



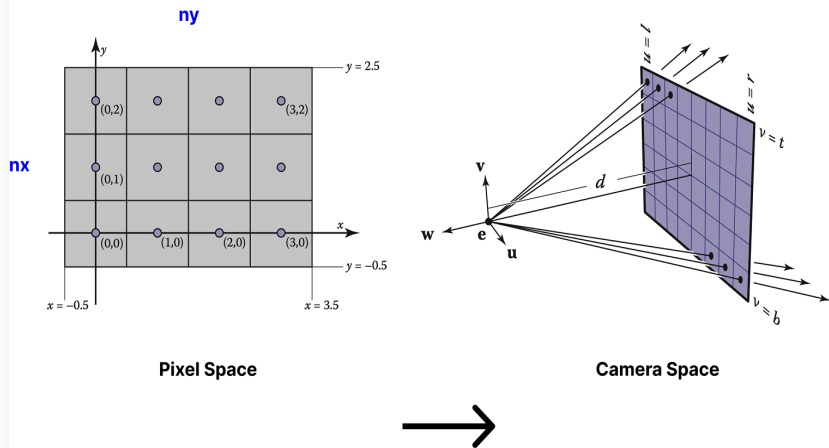
$$\vec{p}(t) = \vec{e} + t\vec{d}$$

$$\vec{e} = ?$$

$$\vec{d} = ?$$

Perspective Projection

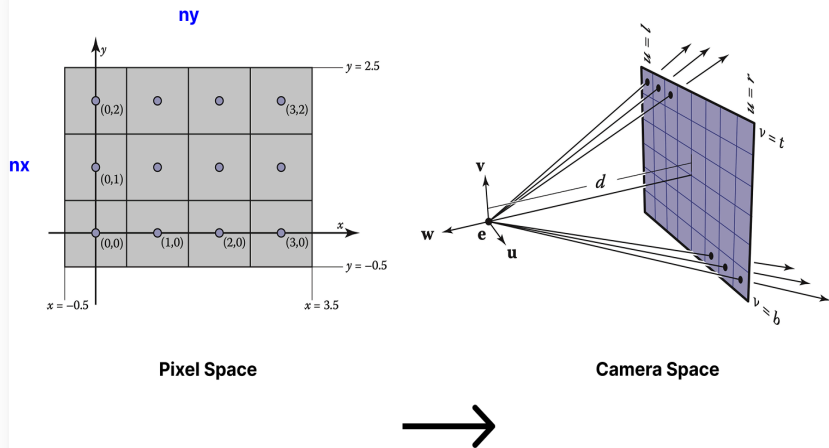
Same origin, different direction.



$$\vec{p}(t) = \vec{e} + t\vec{d}$$
$$\vec{e} = (0, 0, 0)$$
$$\vec{d} = ?$$

Perspective Projection

Same origin, different direction.



$$\vec{p}(t) = \vec{e} + t\vec{d}$$

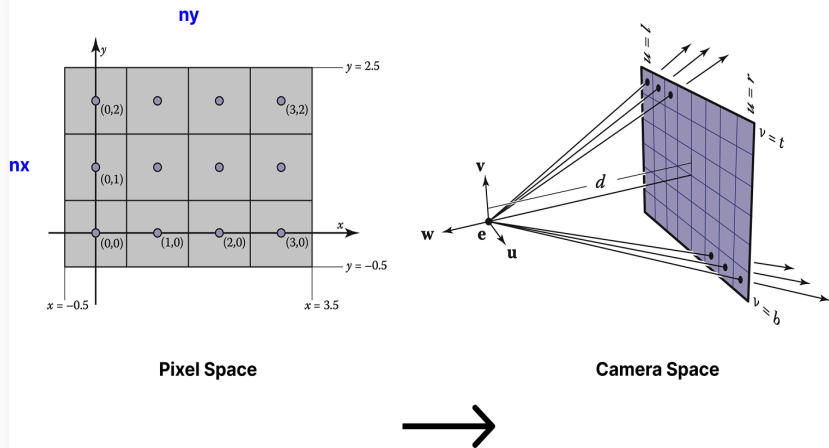
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$$u = \frac{width}{nx}(i + 0.5) - \frac{width}{2}$$
$$v = \frac{height}{ny}(j + 0.5) - \frac{height}{2}$$

Perspective Projection

Same origin, different direction.



```
1  for each pixel {
2      e = (0, 0, 0)
3      u = width/nx * (i + 0.5) - width/2
4      v = height/ny * (j + 0.5) - height/2
5      d = (u, v, focal_length)
6
7      for each object {
8          if (ray intersects object) {
9              Do something
10         }
11     }
12 }
```

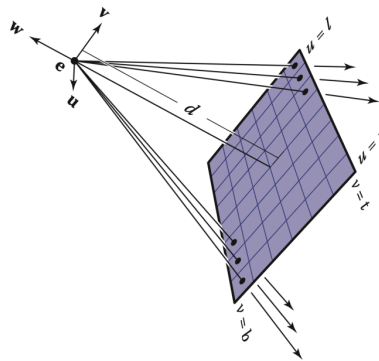
Demo

Ray Equation in Camera Space

General form:

$$\vec{p}(t) = \vec{e} + t(u(i)\vec{u} + v(j)\vec{v} - d\vec{w})$$

$$\vec{p}(t) = \vec{e} + t \begin{bmatrix} \vec{u} & \vec{v} & \vec{w} \end{bmatrix} \begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix}$$



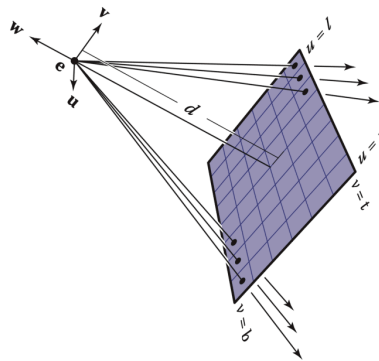
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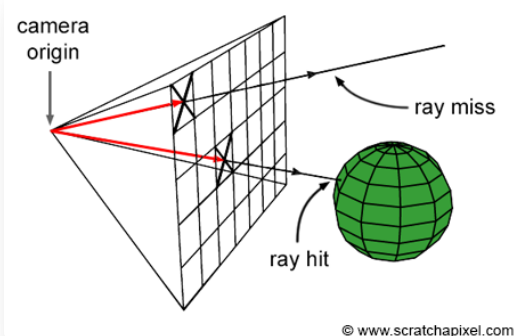
$$\vec{p}(t) = \vec{e} + t \begin{bmatrix} \vec{u} & \vec{v} & \vec{w} \end{bmatrix} \begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix}$$

$\begin{bmatrix} \vec{u} & \vec{v} & \vec{w} \end{bmatrix}$ Is a camera matrix



Ray Casting Mechanisms

```
1  for each pixel {  
2    generate a Ray  
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```



- ☒ ~~x~~ What am I shooting? A Ray: $p(t) = e + t \cdot d$
- ☒ ~~x~~ Where is the ray being shot? From the camera towards the image plane.
- ☐ How do I know if I hit something?

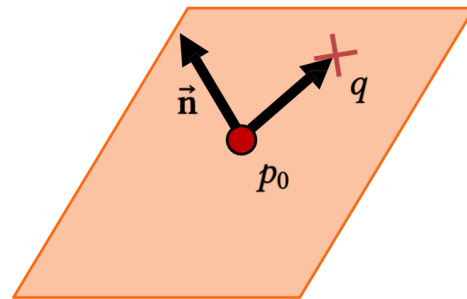
Intersection Test

- Plane
- Sphere
- Triangle

Plane

Intersection with a plane.

$$\vec{n} \cdot (\vec{q} - \vec{p}_0) = 0$$

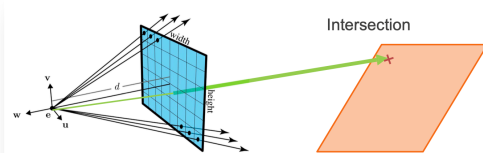


Plane

Intersection with a plane.

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$$\vec{p}(t) = \vec{e} + t\vec{d}$$



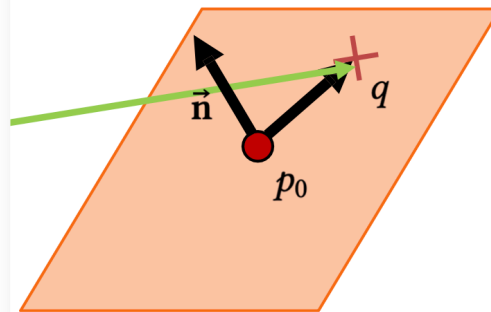
Plane

Intersection with a plane.

$$\vec{n} \cdot (\vec{q} - \vec{p}_0) = 0$$

$$\vec{p}(t) = \vec{e} + t\vec{d}$$

$$\vec{n} \cdot ((\vec{e} + t\vec{d}) - \vec{p}_0) = 0$$



Plane

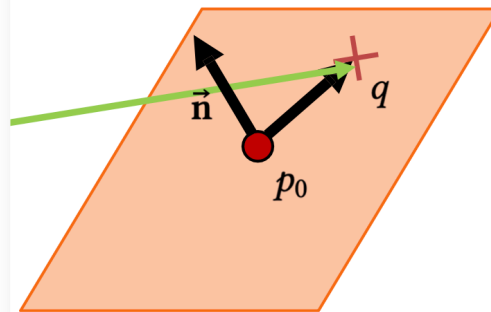
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$$\vec{p}(t) = \vec{e} + t\vec{d}$$

$$\vec{n} \cdot ((\vec{e} + t\vec{d}) - \vec{p}_0) = 0$$

$$\vec{n} \cdot (\vec{e} - \vec{p}_0) + t(\vec{n} \cdot \vec{d}) = 0$$



Plane

Intersection with a plane.

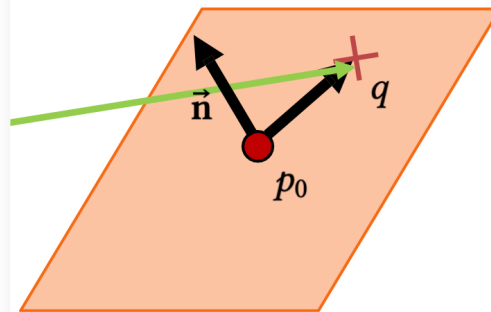
$$\vec{n} \cdot (\vec{q} - \vec{p}_0) = 0$$

$$\vec{p}(t) = \vec{e} + t\vec{d}$$

$$\vec{n} \cdot ((\vec{e} + t\vec{d}) - \vec{p}_0) = 0$$

$$\vec{n} \cdot (\vec{e} - \vec{p}_0) + t(\vec{n} \cdot \vec{d}) = 0$$

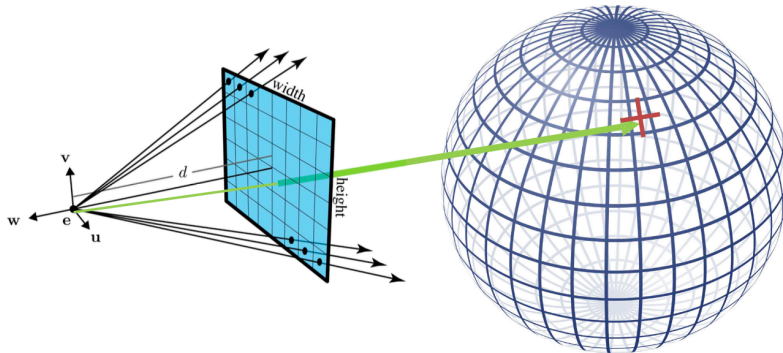
$$t = \frac{\vec{n} \cdot (\vec{p}_0 - \vec{e})}{\vec{n} \cdot \vec{d}}$$



Sphere

Intersection with a Sphere.

A sphere can be defined implicitly with a center $c = (x_c, y_c, z_c)$ and a radius r :

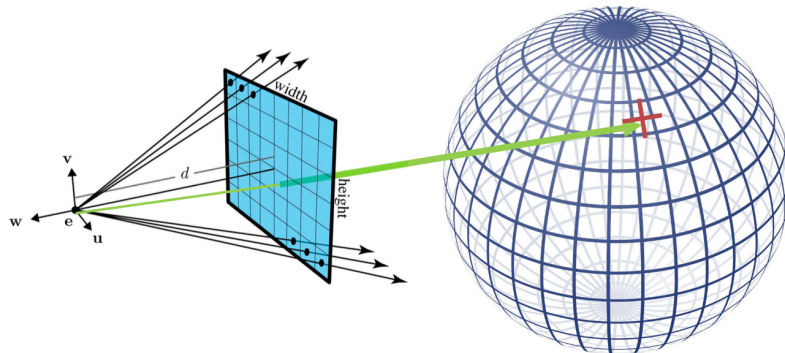


Sphere

Intersection with a Sphere.

A sphere can be defined implicitly with a center $c = (x_c, y_c, z_c)$ and a radius r :

$$(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - r^2 = 0$$



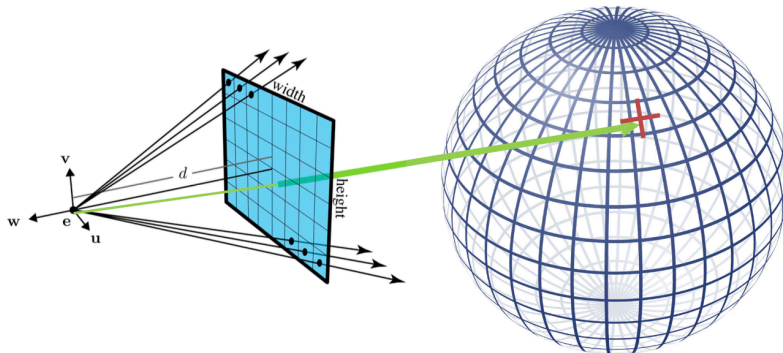
Sphere

Intersection with a Sphere.

A sphere can be defined implicitly with a center $\mathbf{c} = (x_c, y_c, z_c)$ and a radius r :

$$(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - r^2 = 0$$

$$(\vec{p} - \vec{c}) \cdot (\vec{p} - \vec{c}) - r^2 = 0$$



Sphere

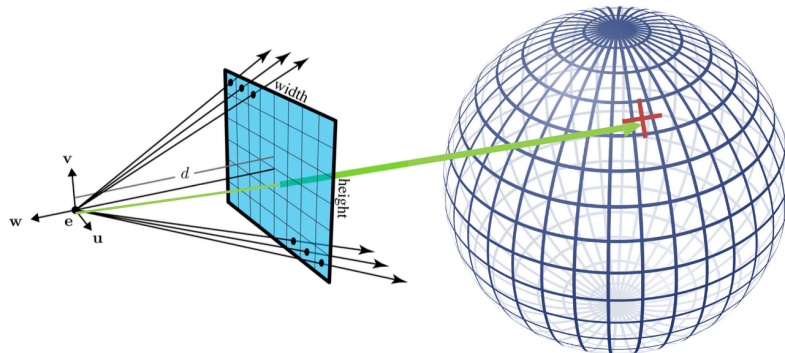
Intersection with a Sphere.

A sphere can be defined implicitly with a center $\mathbf{c} = (x_c, y_c, z_c)$ and a radius r :

$$(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - r^2 = 0$$

$$(\vec{p} - \vec{c}) \cdot (\vec{p} - \vec{c}) - r^2 = 0$$

$$(\vec{e} + t\vec{d} - \vec{c}) \cdot (\vec{e} + t\vec{d} - \vec{c}) - r^2 = 0$$



Sphere

Intersection with a Sphere.

A sphere can be defined implicitly with a center $\mathbf{c} = (x_c, y_c, z_c)$ and a radius r :

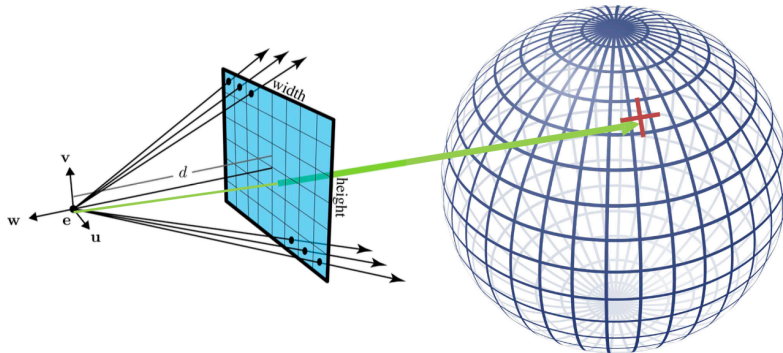
$$(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - r^2 = 0$$

$$(\vec{p} - \vec{c}) \cdot (\vec{p} - \vec{c}) - r^2 = 0$$

$$(\vec{e} + t\vec{d} - \vec{c}) \cdot (\vec{e} + t\vec{d} - \vec{c}) - r^2 = 0$$

Rearranging terms as:

$$(\vec{d} \cdot \vec{d})t^2 + 2\vec{d} \cdot (\vec{e} - \vec{c})t + (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - r^2 = 0$$



Sphere

Intersection with a Sphere.

A sphere can be defined implicitly with a center $c = (x_c, y_c, z_c)$ and a radius r :

$$(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - r^2 = 0$$

$$(\vec{p} - \vec{c}) \cdot (\vec{p} - \vec{c}) - r^2 = 0$$

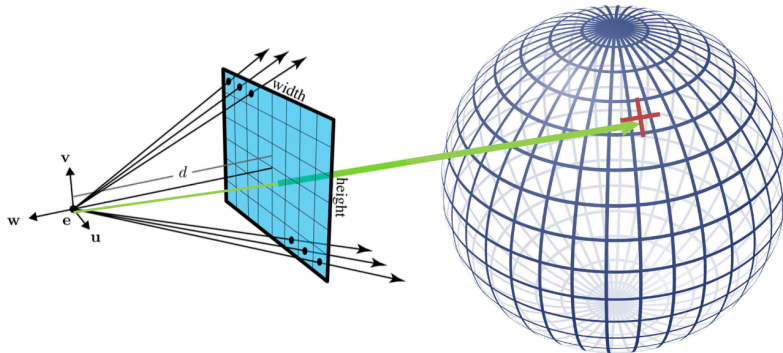
$$(\vec{e} + t\vec{d} - \vec{c}) \cdot (\vec{e} + t\vec{d} - \vec{c}) - r^2 = 0$$

Rearranging terms as:

$$(\vec{d} \cdot \vec{d})t^2 + 2\vec{d} \cdot (\vec{e} - \vec{c})t + (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - r^2 = 0$$

This looks familiar to us, it's a quadratic equation!

$$At^2 + Bt + C = 0$$



Sphere

Intersection with a Sphere.

A sphere can be defined implicitly with a center $c = (x_c, y_c, z_c)$ and a radius r :

$$(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 - r^2 = 0$$

$$(\vec{p} - \vec{c}) \cdot (\vec{p} - \vec{c}) - r^2 = 0$$

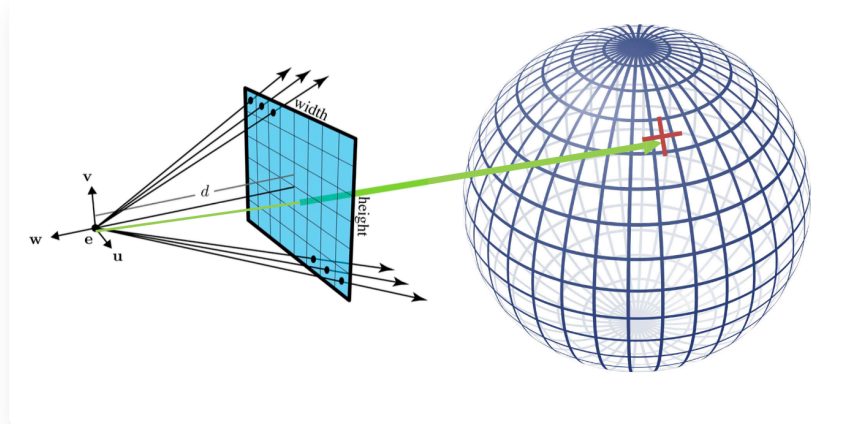
$$(\vec{e} + t\vec{d} - \vec{c}) \cdot (\vec{e} + t\vec{d} - \vec{c}) - r^2 = 0$$

Rearranging terms as:

$$(\vec{d} \cdot \vec{d})t^2 + 2\vec{d} \cdot (\vec{e} - \vec{c})t + (\vec{e} - \vec{c}) \cdot (\vec{e} - \vec{c}) - r^2 = 0$$

This looks familiar to us, it's a quadratic equation!

$$At^2 + Bt + C = 0$$

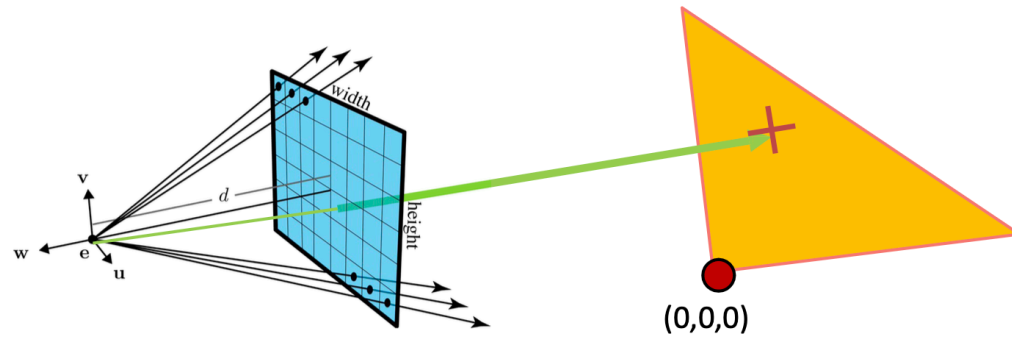


Looking at the discriminant, we can determine if we have an intersection:

- < 0 : no intersection
- $= 0$: one intersection (tangent)
- > 0 : two intersections

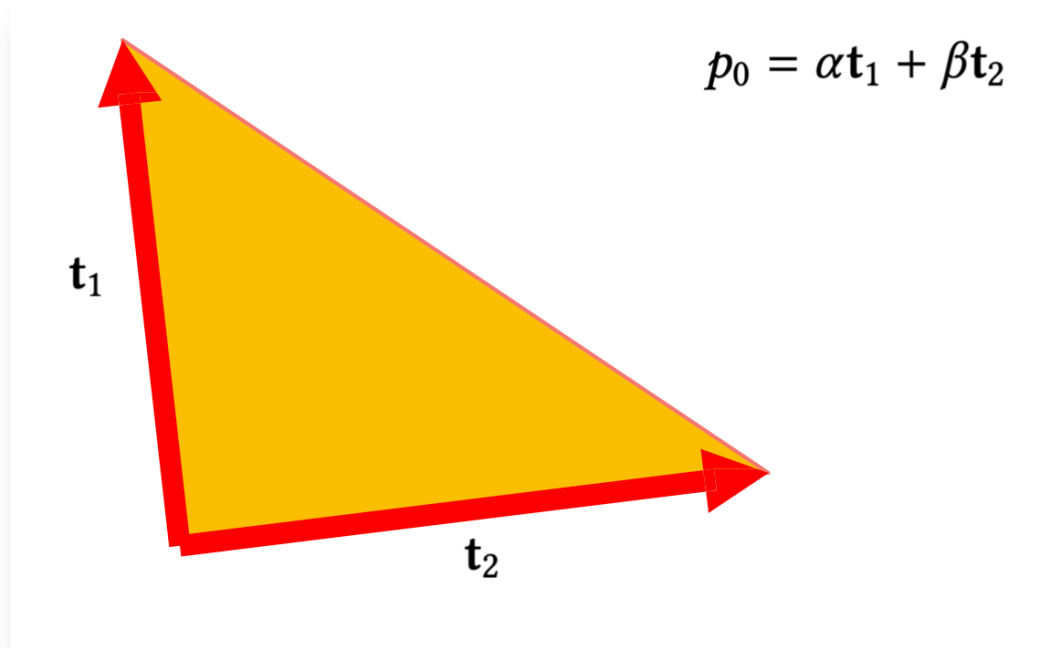
Triangle

Intersection with a Triangle.



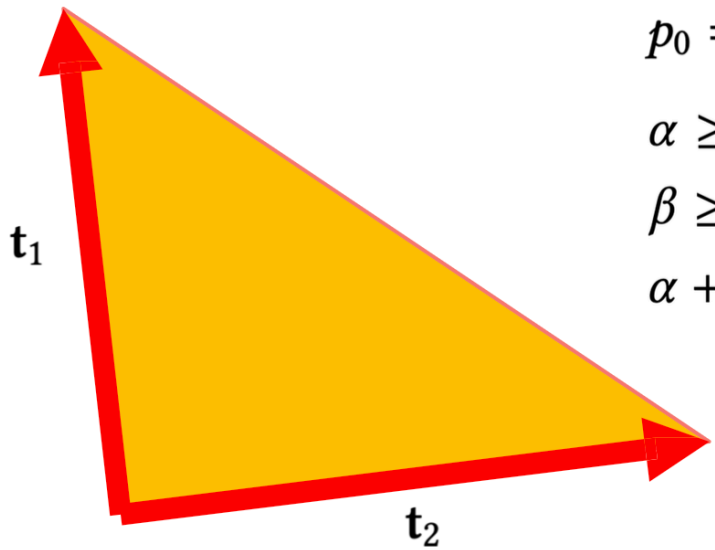
Triangle

Intersection with a Triangle.



Triangle

Intersection with a Triangle.



$$p_0 = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2$$

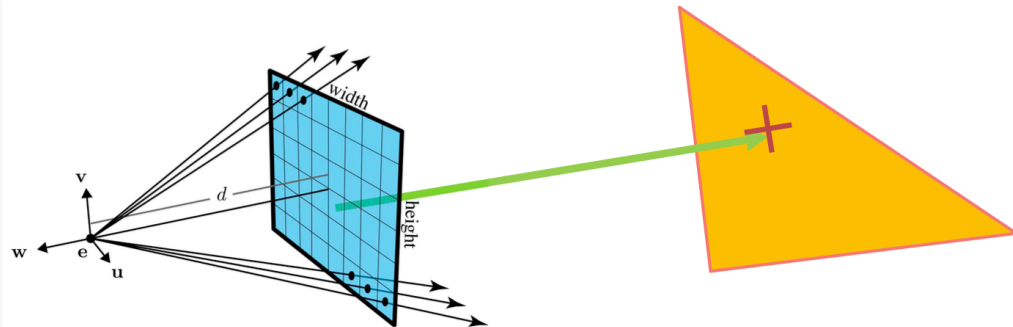
$$\alpha \geq 0$$

$$\beta \geq 0$$

$$\alpha + \beta \leq 1$$

Triangle

Intersection with a Triangle.



Triangle

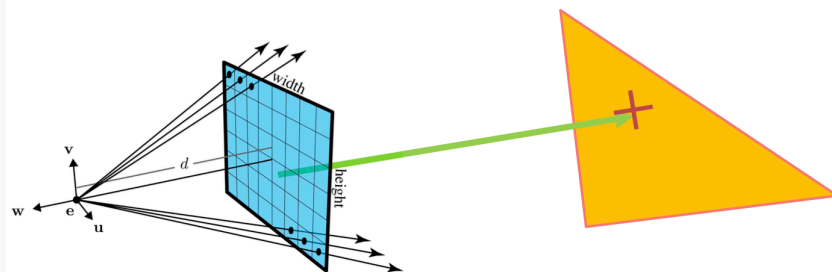
Intersection with a Triangle.

$$\mathbf{p}(t) = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2$$

$$\mathbf{e} + t\vec{\mathbf{d}} = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2$$

$$\mathbf{e} = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2 - t\vec{\mathbf{d}}$$

Check via equating point on surface with point on ray



Triangle

Intersection with a Triangle.

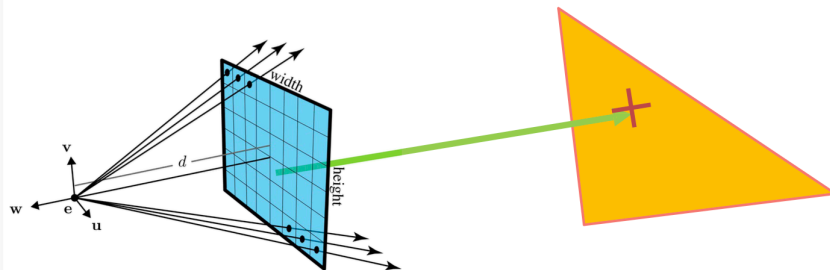
$$\mathbf{p}(t) = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2$$

$$\mathbf{e} + t\vec{\mathbf{d}} = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2$$

$$\mathbf{e} = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2 - t\vec{\mathbf{d}}$$

Check values of α and β

Check via equating point on surface with point on ray



Ray Casting Mechanisms

So what are some mechanisms we need to understand?

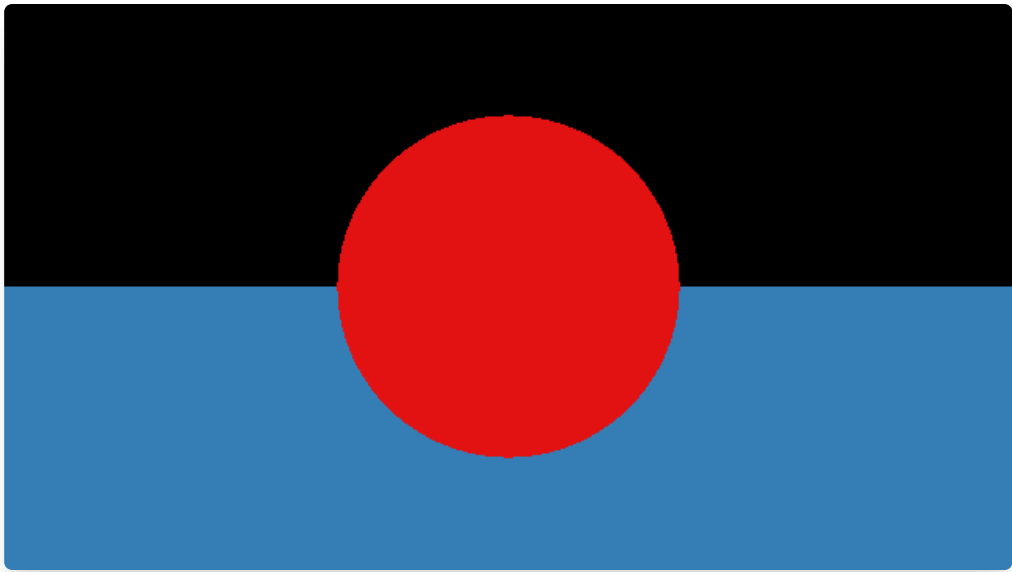
- ☒ ~~x~~ What am I shooting? A Ray: $p(t) = e + t \cdot d$
- ☒ ~~x~~ Where is the ray being shot? From the camera towards the image plane.
- ☒ ~~x~~ How do I know if I hit something? Intersection tests!

```
1  for each pixel {
2      ray = generateRay(i, j)
3      for each object {
4          if (intersects(ray, object)) {
5              setPixel(i, j, object.color)
6          }
7      }
8  }
```

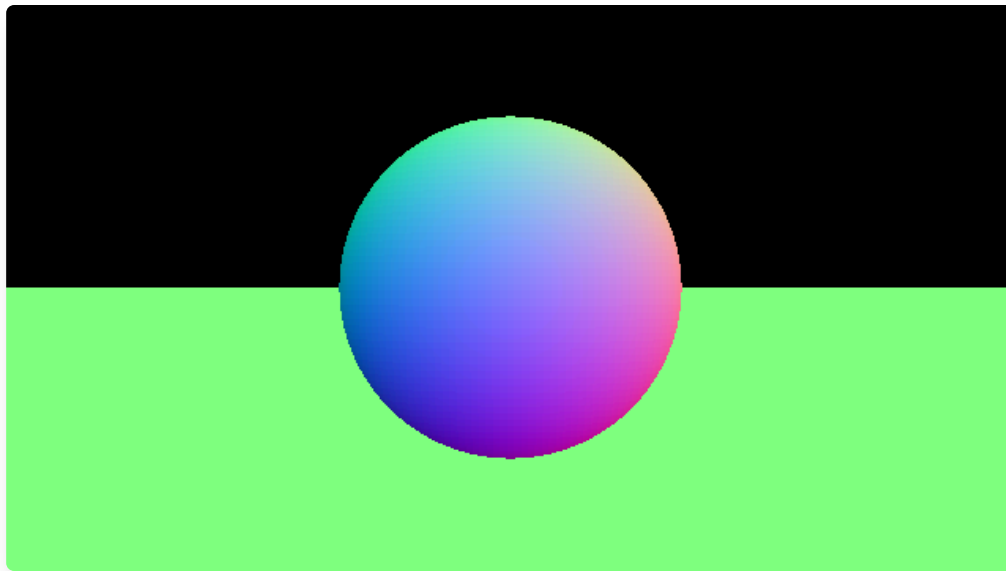
Output Types

- Object ID
- Surface Normal
- Depth

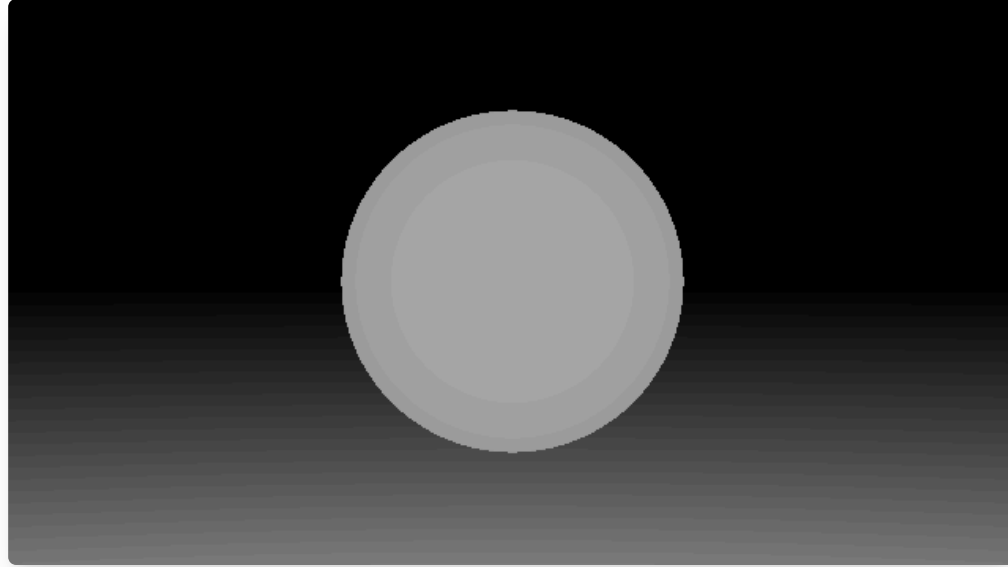
Object ID



Surface Normals



Depth



A2: Ray Casting

Check the course github for links to assignments: <https://github.com/karansher/computer-graphics-csc317>

Quiz 1 will be released tomorrow.